# Ram Moment Calculus for Four-State Potts Reconstruction and Nonlinear Tree Capacity

Source Audit, Uniform Local Bounds, and a Conditional Locally Finite Extension

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**Abstract**

The assigned October 5, 2026 manuscript characterizes reconstruction for the ferromagnetic four-state broadcast process on bounded-degree deterministic trees by nonlinear $L^3$ capacity with resistances $\lambda^{-2|e|}$. We audit its complete proof, preserving the normalization $m = 4\mathbb{E}|p - \mathsf{u}|^2$, the correction $m + Km^3$, the signed third moment, the vertex-dependent terminal noise, and the flow-path accumulation. Its essential preserved-class inequality is traced to the original companion's exact polynomial certificate, whose full two-variable traversals are not independently replayed here. Accordingly, the global conclusions in this report explicitly retain that input as a premise.

The concrete Ram correspondence is exact joint-moment contraction on the posterior tetrahedron, coupled to finite polynomial coordinate conversion and complete rational remainders, not an electrical-network reinterpretation. We prove an unconditional strengthening of the local product estimates: the quadratic-loss constant improves from seven to five, and the third-moment bound has an absolute constant without channel smoothing. A merge-by-merge cubic potential then removes the uniform offspring bound from the necessity argument. With the same companion premise, we prove the two-sided capacity criterion on every locally finite deterministic tree with known, heterogeneous edge strengths $0 < \lambda_e \le \ell < 1$, without a positive lower bound on those strengths. Explicit capacity-to-advantage bounds and a sharper terminal-noise loss are included. A finite-capacity recursion, its residual budget, and critical examples distinguish genuine infinite-tree conclusions from finite diagnostics. No independent certification of the companion, historical novelty, or measured computational acceleration is claimed. Contribution assessment: C3/5, uniform local estimates and a proved conditional extension to locally finite heterogeneous trees are notable advances relative to the assigned source, while the companion certificate remains an external premise.

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# 1 Problem, provenance, and claim boundaries

## 1.1 The exact assigned theorem

The mathematical target is OpenAI, *A Capacity Criterion for Four-State Potts Reconstruction on Trees*, dated October 5, 2026 [1]. Its complete `build/main.tex`, including all proofs, was read. The selected public pin is

`adc7f1241b42e322a6451854ab7e4b4c146bf78a`.

The existing October 8 upstream identity audit reports no changed selected source at commit `fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb`. That provenance result is reused, not rerun, and is not a proof-validation result.

Let $T$ be an infinite deterministic rooted tree, with root $\rho$ and at most $B < \infty$ children at each vertex. Leaves and one-child vertices are allowed. The whole tree is known to the observer. Fix $0 < \lambda < 1$. The root spin is uniform on [4]; independently along edges conditional on parent spins, use

$$P_\lambda(j \mid i) = \lambda \mathbf{1}_{\{i=j\}} + (1-\lambda)/4. \tag{1.1}$$

Thus $\lambda$ is the nonconstant eigenvalue, while the probability of copying the spin is $(1+3\lambda)/4$. Write $L_n = \{v : |v| = n\}$ and $\mathsf{u} = (1/4, 1/4, 1/4, 1/4)$. With expectation over broadcast randomness on this fixed tree, define

$$a_n = \mathbb{E}\,\mathrm{TV}\big(\mathbb{P}(\sigma_\rho \in \cdot \mid \sigma_{L_n}), \mathsf{u}\big), \qquad \mathrm{TV}(p, \mathsf{u}) = \tfrac{1}{2}\sum_i |p_i - 1/4|. \tag{1.2}$$

Data processing gives $a_{n+1} \le a_n$; reconstruction means $a_\infty = \lim_n a_n > 0$. An empty observed level is uninformative. No averaging over a random tree or conditioning on survival is hidden in this definition.

A nonnegative flow $\theta$ to infinity obeys conservation at every nonroot vertex, including leaves. Its root mass is $|\theta| = \sum_{w \in \mathrm{ch}(\rho)} \theta(\rho, w)$. At a finite leaf conservation forces incoming flow to be zero. With $|e|$ the depth of the child endpoint and $\partial T$ the infinite rays, the exact capacity is

$$V_\lambda(\theta, \xi) = \sum_{e \in \xi} (\lambda^{-2|e|}\theta(e))^2, \tag{1.3}$$

$$\mathrm{Cap}_{3,\lambda}(T) = \sup\{|\theta| : \theta \ge 0 \text{ is a flow to infinity}, \sup_{\xi \in \partial T} V_\lambda(\theta, \xi) \le 1\}. \tag{1.4}$$

The source's Theorem 1.1 asserts $a_\infty > 0$ if and only if $\mathrm{Cap}_{3,\lambda}(T) > 0$. The exponent two in the *path potential* means $p - 1 = 2$, hence $p = 3$. The resistance is separately $\lambda^{-2|e|}$. Replacing this by ordinary electrical $L^2$ capacity changes the theorem. At $\lambda = 1$, an infinite ray reconstructs perfectly but the capacity in (1.4) vanishes. This endpoint is genuinely excluded.

## 1.2 Evidence classes and the realized contribution

| Status | Content |
|---|---|
| Inherited source | Bounded-degree homogeneous capacity equivalence; posterior algebra, corrected labels, flow construction, noisy-boundary comparison and path summation. |
| External premise | The preserved-class moment-saving statement MS below. The companion asserts an exact proof; this report does not independently certify its complete traversals. |
| Unconditional local results | The uniform product bounds in Theorem 5.2, including zero denominators, and the finite-capacity results in Section 8. |
| Conditional extensions proved here | Degree-independent corrected-moment subadditivity, arbitrary locally finite known trees with heterogeneous strengths bounded above by $\ell < 1$, and quantitative advantage bounds. |
| Finite executed diagnostics | Rational identities and small posterior, flow, and recursion fixtures. They do not certify MS or an arbitrary infinite tree. |
| Not claimed | A new unconditional Potts theorem, a full independent companion replay, a general posterior moment closure, or a numerical speedup. |

Catalogue level L4 assesses methodological relevance. It is distinct from the C1–C5 realized-contribution scale: C1 means none demonstrated, C2 a little, C3 notable, C4 substantial/high, and C5 very high. C3 is assigned here for the actual conditional domain extension and unconditional local strengthening, not for report length, inherited capacity theory, or future certificate work.

### 1.3 Current primary literature

Public searches on October 8, 2026 used only public mathematical titles and terms. Pemantle–Peres [3] give the classical nonlinear-capacity framework, including $L^2$ free-boundary and $L^3$ plus-boundary Ising criteria with different resistance conventions. Their capacity recursion supplies relevant precedent; an Ising boundary condition is not the four-state posterior experiment. Sly [4] treats symmetric Potts reconstruction, with large-degree three-state sharpness and nonsharpness for at least five states. Mossel–Sly–Sohn [5], Theorem 1.12, prove three-/four-state nonreconstruction including equality for sufficiently large mean degree under their uniform-tail and tightness assumptions. That is not the assigned arbitrary-tree theorem. Janson–Mossel [6] provide the robust-reconstruction context; the present flow-calibrated noise varies across vertices and observation depths and does not assert an arbitrary-noise robustness theorem. The public source itself was located in the current repository. This bounded literature check is not an exhaustive priority search.

## 2 Posterior moment calculus and the companion contract

### 2.1 Coordinates, support, and reference measures

Let $\Delta_4 = \{p \in [0,1]^4 : \sum_i p_i = 1\}$ and set

$$L = \begin{pmatrix} 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}, \qquad x = Lp, \qquad p = \mathsf{u} + \tfrac{1}{4} L^{\mathsf{T}} x. \tag{2.1}$$

The domain is the tetrahedron $L\Delta_4$, not the entire cube $[-1,1]^3$. A posterior law $\mu$ is the law of $p(Y)$ under the marginal observation law. Symmetry means invariance under all color permutations. Define

$$A = |x|^2 = 4|p - \mathsf{u}|^2, \quad m(\mu) = \mathbb{E}_\mu A, \quad \tau(\mu) = \mathbb{E}_\mu x_1 x_2 x_3, \quad h(\mu) = |\tau(\mu)|. \tag{2.2}$$

Crucially $h$ is the absolute value *after* the signed expectation, not $\mathbb{E}|x_1x_2x_3|$. Since $L^\mathsf{T}L = 4I - \mathbf{1}\mathbf{1}^\mathsf{T}$,

$$0 \le m \le 3, \qquad m/8 \le \mathbb{E}\,\mathrm{TV}(p, \mathsf{u}) \le \sqrt{m}/2. \tag{2.3}$$

The lower bound follows from $\sum_i (p_i - 1/4)^2 \le \sum_i |p_i - 1/4|$ and the upper from Cauchy–Schwarz and Jensen. The constants are adequate, not optimal. Degrading an observation replaces its posterior by a conditional expectation; convexity of $A$ proves monotonicity of $m$.

For a channel step $P_t$, $0 \le t \le 1$, the marginal observation law stays the same and

$$x \mapsto tx, \qquad m \mapsto t^2 m, \quad \tau \mapsto t^3 \tau, \quad h \mapsto t^3 h. \tag{2.4}$$

For two observations independent conditional on their uniform input, sample $p, q$ independently from their *marginal* posterior laws. Then

$$Z = 4\sum_i p_i q_i, \quad P_i = 4p_i q_i / Z, \qquad \mathbb{E}_{\mu\star\nu} f = \mathbb{E}_{\mu\otimes\nu}[Z f(P)]. \tag{2.5}$$

At $Z = 0$ the weighted integrand is zero. The formula follows from the conditional likelihoods $4p_i$ and $4q_i$, averaged over the four input colors. The barycenters are $\mathsf{u}$, so $\mathbb{E}Z = 1$. Factorization is valid before the tilt by $Z$, not under the actual combined observation law.

Color permutations act on $x$ by coordinate permutations and even sign changes. Consequently

$$\mathbb{E}x_i = 0, \qquad \mathbb{E}x_i x_j = (m/3)\mathbf{1}_{i=j}, \tag{2.6}$$

and the only possibly nonzero degree-three coordinate moment is $\tau$. This symmetry is not Gaussianity: $\tau$ need not vanish. Every coordinate monomial of degree at least two has absolute expectation at most $m$, because $|x_i| \le 1$ and two retained factors are bounded by $|x|^2$.

## 2.2 What MS says, and where it comes from

**Premise MS.** There exists a class $\mathcal{C}$ of symmetric posterior laws containing the revealing and uninformative laws, closed under all channel steps $0 \le t \le 1$ and conditionally independent products, such that for every $\mu, \nu \in \mathcal{C}$,

$$m(\mu \star \nu) \le a + b - c\{\mathbb{E}_\mu A^2\, b + a\mathbb{E}_\nu A^2\}, \qquad a = m(\mu), \quad b = m(\nu), \quad c = 1/1000. \tag{2.7}$$

This is exactly the source's `prop:companion`, not an assumption that symmetry alone implies subadditivity. The original companion [2], Section 3, `prop:cone-closure`, actually has the needed all-$t$, unequal-input product scope. It defines $\mathcal{C}$ by $\mathbb{E}H \ge 0$, where, putting $J = x_1x_2x_3$ and $C = x_1^2x_2^2 + x_1^2x_3^2 + x_2^2x_3^2$,

$$\begin{aligned} H &= J - A^2/10 - 3C/20 + 3AJ/10 - A^3/50 + 7AC/100 - 51J^2/100, \\ F &= 4A - 3J + 7A^2/10 - 4C - 4AJ/5, \\ G &= 1 - 161A/100 - 31J/10 + 773A^2/1000 + 4C - 199AJ/50. \end{aligned} \tag{2.8}$$

The renamed invariant $J$ avoids confusing it with the tree bound $B$. The companion's `prop:polynomial-inequalities` asserts $F, G \geq 0$, $H(tx) \geq t^3 H(x)$ and, for every $p, q \in \Delta_4$,

$$[ZA(P)] \leq A(p) + A(q) - H(p)F(q) - F(p)H(q) - cA(p)A(q)(A(p) + A(q)), \tag{2.9}$$

$$[ZH(P)] \geq H(p)G(q) + G(p)H(q). \tag{2.10}$$

Brackets average the weighted expression over all 24 permutations of $q$. Integrating (2.10) against independent marginal laws proves preservation of $\mathbb{E}H \geq 0$. Integrating (2.9) and dropping the nonnegative factored $H, F$ terms proves (2.7). Revealing and uninformative laws qualify since $H(e_i) = 13/100$ and $H(\mathtt{u}) = 0$. The condition is on the law, not individual posteriors: $H(1/2, 1/2, 0, 0) = -3/25$.

Original Sections 5 and 6 reduce these claims to scalar signs and product-simplex rational inequalities, then to exact Type T or Type P coefficient tests on a midpoint subdivision. Type T uses a fifth-order Taylor minorant for $z^{-k}$ multiplied by a separately nonnegative numerator correction. Type P pays for the remainder of its actual degree-$(3, 3)$ interpolant through exact coefficient correction; it is not a node-sampling proof. The continuous weighted terms $z\phi(P)$ extend by zero when $z = 0$. Both children at every midpoint split cover their parent, including equality faces. The source reports successful full traversals with 4479 and 1935 visited cells and no unresolved leaves. Those counts are *source-reported*, not executions in this report.

The related local report M437 [7] explicitly leaves these full traversals unreplayed while recording independent scalar certificates and conditional consequences. It is useful corroborating analysis but does not replace the original contract or turn MS into an independently verified theorem here. The companion's statement itself is unconditional; the conditional language used here describes this report's evidence boundary, not a claim that the published statement contains an explicit conditional hypothesis. No counterexample to MS is asserted.

At every finite depth in the assigned proof, all subtree laws and partial products belong to $\mathcal{C}$ under MS. Individually selected terminal channels also qualify. No limiting posterior law is required to belong to $\mathcal{C}$: only scalar limits of finite-depth inequalities are used.

# 3 Detailed audit of the assigned capacity proof

## 3.1 Necessity: the correction must precede the flow

Set $r = \lambda^2$. Let $m_v^{(n)}$ be the information about $\sigma_v$ from descendants at distance $n$, initially $m_v^{(0)} = 3$. Data processing yields $m_v^{(n)} \downarrow m_v \in [0, 3]$. If the root reconstructs then $m_\rho > 0$ by (2.3). For $b_w = r m_w$ put

$$S = \sum_w b_w, \qquad D = \max_{w \neq w'} b_w b_{w'} (b_w + b_{w'}),$$

with $D = 0$ for fewer than two children. Combining a selected pair first, using Jensen in (2.7), then subadditivity for the others, gives

$$m_v \leq S - cD, \qquad S^3 \leq \sum_w b_w^3 + B^3 D. \tag{3.1}$$

The second inequality bounds each mixed ordered triple by the largest child value squared times the second largest, hence by $D$. There are at most $B^3$ such triples. Finite child sets justify the limit and the maximum.

Choose $0 < K \le c/B^3$ and $y_v = m_v + Km_v^3$. The original derivation is

$$y_v \le S - cD + KS^3 \le r\sum_w y_w - Kr(1-r^2)\sum_w m_w^3. \tag{3.2}$$

Since $y_w \le (1+9K)m_w$, define $\delta = K(1-r^2)/(1+9K)^3 > 0$. The tangent inequality $1-t \le (1+2t)^{-1/2}$ gives

$$y_v \le r\sum_w y_w(1-\delta y_w^2) \le r\sum_w \frac{y_w}{\sqrt{1+2\delta y_w^2}}.$$

Thus $z_v = \sqrt{2\delta}\, y_v$ satisfies $z_v \le \sum_w r z_w/\sqrt{1+z_w^2}$. Without the cubic correction, a one-child vertex has only $m_v = rm_w$, with no pair saving. The term $r - r^3$ supplies the missing strict attenuation in the corrected variable.

The source's flow lemma chooses $0 \le d_{vw} \le r/\sqrt{1+z_w^2}$ with $z_v = \sum_w d_{vw} z_w$. Taking $l_\rho = 1$, $l_w = l_v d_{vw}$, and $\theta(v,w) = l_w z_w$ gives conservation. For $g_v = r^{-|v|} l_v$,

$$g_w^2(1+z_w^2) \le g_v^2, \quad (r^{-|w|}\theta(v,w))^2 \le g_v^2 - g_w^2. \tag{3.3}$$

Every finite path sum is at most one; monotone convergence proves the bound on every infinite ray. The root mass is $z_\rho > 0$. This is an explicit nonlinear flow construction, not the inversion of a graph Laplacian.

### 3.2 Local moments: the signed cancellation

For independent marginal coordinate vectors $x, y$, put $s = x \cdot y$ and

$$Z = 1 + s, \quad N = x + y + R, \quad R_i = x_j y_k + x_k y_j \quad (\{i,j,k\} = \{1,2,3\}), \quad LP = N/Z. \tag{3.4}$$

Writing $a = m(\mu)$, $b = m(\nu)$, $\alpha = \tau(\mu)$ and $\beta = \tau(\nu)$, symmetry gives

$$\mathbb{E}|R|^2 = 2ab/3, \quad \mathbb{E}s^2 = ab/3, \quad \mathbb{E}[sx \cdot R] = 2\alpha b, \quad \mathbb{E}[sy \cdot R] = 2a\beta, \quad \mathbb{E}[s|R|^2] = 6\alpha\beta.$$

The $ab$ terms cancel exactly, yielding the source identity

$$m(\mu \star \nu) = a + b - 4(\alpha b + a\beta) - 6\alpha\beta + \mathbb{E}\frac{|N|^2 s^2}{1+s}. \tag{3.5}$$

The last term is nonnegative. Replacing $\tau$ by an unsigned third absolute moment *before* expansion obscures this identity. The source obtains its lower bound with constant seven using $h \le m$. For the third moment it expands $\mathbb{E}[N_1N_2N_3/(1+s)^2]$, bounds the exact denominator remainder, and controls mixed monomials using (2.6). Its assumption that one input is channel-smoothed gives $1 + s \ge 1 - \lambda > 0$. Section 5 strengthens both steps without that denominator assumption.

### 3.3 Sufficiency: terminal noise and accumulated flow loss

Take an admissible flow $\theta$ of positive finite mass, scale it by $\epsilon \in (0,1]$, and let $\Theta(v)$ be its entering mass, with $\Theta(\rho) = M = \epsilon|\theta|$. Define

$$X_v = r^{-|v|}\Theta(v), \qquad X_v = r\sum_w X_w. \tag{3.6}$$

Every positive-flow edge continues to an infinite ray, so $X_v \le \epsilon$ away from the root. The root can be larger: the source correctly uses

$$\sum_{v\in\xi,\, v \text{ including } \rho} X_v^2 \le Q\epsilon^2, \qquad Q = 1 + (Br)^2. \tag{3.7}$$

At each $v \in L_n$, independently observe $\sigma_v$ through $P_{t_v}$ with $t_v = \sqrt{X_v/3}$. This is a valid degradation and gives terminal information $\widehat{m}_v = X_v$. Subadditivity and (3.6) imply $\widehat{m}_v \le X_v$ throughout the finite tree. Zero-flow or unobserved subtrees are consistent with this induction.

For $|v| < n$ put $D_v = r\sum_w \widehat{m}_w - \widehat{m}_v$. Sequential products and the local bounds give, for a fixed constant $C_1$,

$$\widehat{h}_v \le \lambda^3 \sum_w \widehat{h}_w + C_1 X_v^2, \tag{3.8}$$

$$0 \le D_v \le C_1 X_v(\lambda^3 \sum_w \widehat{h}_w + X_v^2). \tag{3.9}$$

The propagated third moment has coefficient *exactly* $\lambda^3$, not $C_1\lambda^3$. Iteration inserts the error constant once per internal site; it does not raise that constant to the tree depth. Telescoping gives

$$\widehat{m}_\rho = M - \sum_{|v|<n} r^{|v|} D_v. \tag{3.10}$$

Sampling a terminal vertex with probabilities $\Theta(v)/M$ yields its ancestor path $(V_i)_{i=0}^n$ with $\mathbb{P}(V_j = u) = \Theta(u)/M$. The decisive cancellation of weights is

$$r^i \lambda^{3(j-i)} X_u = \lambda^{j-i}\Theta(u), \qquad |u| = j. \tag{3.11}$$

Internal losses reduce to $M\lambda^{j-i}\mathbb{E}[X_{V_i}X_{V_j}]$, terminal losses to $M\lambda^{n-i}\mathbb{E}X_{V_i}$. Cauchy–Schwarz along each supported path and the geometric series bound the loss divided by $M$ by

$$C_2\left\{\frac{Q\epsilon^2}{1-\lambda} + \frac{\sqrt{Q}\,\epsilon}{1-\lambda}\right\}.$$

One choice of $\epsilon > 0$, independent of $n$, makes this at most $1/2$, hence $a_n \ge M/16$. The degraded experiments may vary with $n$. Neither their convergence nor monotonicity is claimed or needed.

### 3.4 Audit conclusion and exact source locators

No additional gap was identified in the assigned source's analytic steps under MS. Bounded degree enters its necessity estimate through $B^3$ and its flow normalization through the root mass bound. The local and accumulated-loss calculations already suggest ways to avoid a uniform degree hypothesis. The actual source locators are:

| Source block | Labels in assigned `build/main.tex` |
|---|---|
| Theorem/domain, lines 43–111 | `thm:main`, `eq:potential`, `eq:capacity`. |
| Posterior/companion, 193–312 | `eq:statistics`, `eq:product-rule`, `prop:companion`. |

| Source block | Labels in assigned `build/main.tex` |
|---|---|
| Necessity, 314–424 | `lem:flow`, `eq:pair-saving`, `eq:renormalized-saving`. |
| Local proof, 426–537 | `lem:local-moments`, `eq:quadratic-cancellation`, `eq:denominator-remainder`. |
| Sufficiency, 539–758 | `eq:terminal-noise`, `eq:unfolded-loss`, `eq:weight-identity`, `eq:uniform-loss-bound`. |

The SHA-identified read-scope ledger fixes these line locators; labels are the preferred stable references.

# 4 The precise Ram dictionary

## 4.1 Private proofs actually read

The three routing maps and catalogue row 438 identify relevant methods, not proofs of the Potts claim. The following original confidential passages were read directly and used only within their domains. Paths in this subsection start at `C:/Users/mural/OneDrive/Documents/`.

| Original and exact locator | Valid transfer and nontransfer |
|---|---|
| `RamPatsCore50-TRs/TR-core-pat-16/TR-core-pat-16.tex`; `eq:momentcontract`, `prop:psd`, lines 217–254 | A common finite polynomial contracts against moments of one law. The proof is covariance congruence. PSD does not prove support-realizability or the companion inequality. |
| Same file; `prop:gaussian`, `eq:nongaussian`, lines 271–304 | Wick contraction needs a Gaussian law. Its non-Gaussian formula retains third and fourth moments. Here symmetry leaves $\tau$, so a Gaussian replacement is invalid. |
| Same file; `thm:remainder`, `eq:covidentity`, lines 305–334; `prop:rankloss`, lines 448–461 | A remainder is bounded under the same probability law; almost-sure nonzero denominators do not imply inverse-moment integrability. The complete Bayes-weighted expression, not a naked reciprocal, is the relevant object. |
| `RamFoundationTRs/TR-rev-1-old-tr250/TR-rev-1-old-tr250.tex`; `eq:moments`, "What a Ram representation means", lines 155–178; "A concrete polynomial Ram construction", 455–475 | Finite moment coordinates $u = Ec$ are exact only on the named finite space with injective $E$. Positive-weight polynomial Gram matrices are unisolvent; no arbitrary posterior distribution is reconstructed from finitely many moments. |

| Original and exact locator | Valid transfer and nontransfer |
|---|---|
| `RamCalBooks/volumes/volume16/chapters/ch05d-clearance.tex`; `v16c05:conversionproof`, `v16c05:hull`, "Subdivision without introducing approximation", lines 62–140 | Exact Bernstein conversion and the nonnegative partition-of-unity proof certify whole panels. The zero-margin and depth-exhaustion warnings apply to the companion's uniform-message faces. No finite grid proves an all-posterior inequality. |

The full clearance chapter was read; relevant complete proofs in the two reports were read. Unrelated private PDE, control, and neural-inference claims are not imported.

### 4.2 Posterior moments are not certificate-polynomial moments

For $d \geq 0$, use the finite moment functional $\mathcal{M}_\mu(x^\alpha) = \mathbb{E}_\mu x^\alpha$, $|\alpha| \leq d$. Channel transport is exactly diagonal:

$$\mathcal{M}_{t\mu}(x^\alpha) = t^{|\alpha|}\mathcal{M}_\mu(x^\alpha). \tag{4.1}$$

Under independent marginal inputs, polynomial products contract as

$$\mathbb{E}_{\mu\otimes\nu} \sum_{\alpha,\beta} c_{\alpha\beta} x^\alpha y^\beta = \sum_{\alpha,\beta} c_{\alpha\beta} \mathcal{M}_\mu(x^\alpha)\mathcal{M}_\nu(y^\beta). \tag{4.2}$$

This is the source-specific finite joint-moment Ram operation. Applied to $|N|^2(1-s)$ it produces (3.5)'s polynomial part. It does *not* replace $\mathbb{E}[|N|^2/(1+s)]$ by an exact closed recurrence in $(m,\tau)$: its rational remainder remains.

For a certificate polynomial on a reference simplex, instead fix normalized uniform measure $\omega$ on $\Delta_4$. Let $B_I^d(X) = \binom{d}{I} X^I$, $|I| = d$. If $R = \sum_I \beta_I B_I^d$, its reference moments $u_I = \int B_I^d R \,\mathrm{d}\omega$ obey $u = E_d\beta$, where

$$(E_d)_{IK} = \frac{6(d!)^2}{(2d+3)!} \prod_{j=0}^{3} \frac{(I_j + K_j)!}{I_j! K_j!}. \tag{4.3}$$

Indeed $\int X^\alpha \,\mathrm{d}\omega = 6\prod_j \alpha_j!/(|\alpha|+3)!$. For every nonzero coefficient vector, the associated polynomial has strictly positive squared integral, so $E_d$ is positive definite. Tensoring gives the corresponding product-simplex matrix. The sufficient positivity test is $E_d^{-1}u \geq 0$, or its tensor counterpart, *not* $u \geq 0$.

For example $E_1 = (I + \mathbf{1}\mathbf{1}^\mathsf{T})/20$. The polynomial $X_0 - 1/8$ has moments $(11,3,3,3)/160 > 0$, but equals $-1/8$ on $X_0 = 0$. Exact moment synthesis recovers its negative Bernstein coefficients. The local M437 report already develops this finite representation; it is used here as an inherited interface, not credited again as a new result. The private proofs justify its exactness and its limitations.

### 4.3 Why the dictionary is useful but not sufficient

There are three separate objects: a random posterior law, a polynomial certificate against a reference measure, and a flow on an infinite tree. Their moment symbols, normalizations and positivity cones cannot be interchanged. Finite exact polynomial algebra helps verify cancellations and local inequalities. Whole-domain remainder bounds are needed for rational posterior updates. Flow

conservation and path-energy summability are needed for the infinite-depth implication. Neither a PSD covariance nor an invertible finite Gram matrix supplies the latter two obligations.

## 5 Uniform local product estimates

We now strengthen the assigned local lemma without using MS. All laws in this section are genuine symmetric posterior laws on $\Delta_4$.

**Lemma 5.1.** *For every such law, $h \le m/3$.*

*Proof.* Let $q = |x_1x_2x_3| \le 1$. Arithmetic–geometric mean gives $|x|^2/3 \ge q^{2/3} \ge q$. Take expectation and then $|\mathbb{E}x_1x_2x_3| \le \mathbb{E}q$. Equality is possible for the revealing law, where $m = 3$ and $\tau = 1$. □

**Theorem 5.2** (No channel smoothing is required)**.** *For every pair of symmetric posterior laws $\mu, \nu$, including laws with zero product denominators on sets of positive product-marginal measure, put $a = m(\mu)$, $b = m(\nu)$. Then*

$$m(\mu \star \nu) \ge a + b - 5\{h(\mu)b + ah(\nu)\}, \tag{5.1}$$

$$h(\mu \star \nu) \le h(\mu) + h(\nu) + Cab, \qquad C = 460. \tag{5.2}$$

*The numerical value 460 is a convenient explicit universal constant, not an optimized one.*

*Proof.* Use (3.4). For $Z > 0$, $|N|^2 \le 3Z^2$ and $|N_1N_2N_3| \le Z^3$, since $LP$ is in the posterior tetrahedron. At $Z = 0$, each $4p_iq_i = 0$, so $N = 0$. The terms $|N|^2s^2/Z$ and $N_1N_2N_3/Z^2$ extend by zero; they are uniformly bounded on the compact product simplex. Thus the algebraic expansion (3.5) is valid in expectation without any lower bound on $Z$.

With $\alpha = \tau(\mu)$ and $\beta = \tau(\nu)$, Lemma 5.1 yields

$$|\alpha\beta| \le \tfrac{1}{6}(h(\mu)b + ah(\nu)).$$

Dropping the nonnegative remainder in (3.5) gives (5.1): the coefficient is $4 + 6/6 = 5$. No sign is assumed for $\alpha$ or $\beta$.

For the third moment use the exact identity

$$\frac{1}{(1+s)^2} - (1 - 2s) = \frac{s^2(3+2s)}{(1+s)^2}. \tag{5.3}$$

On the simplex $-1 \le s \le 3$. Multiplication by the full numerator gives

$$\left|N_1N_2N_3\left\{(1+s)^{-2} - (1-2s)\right\}\right| \le (1+s)(3+2s)s^2 \le 36s^2.$$

The expression extends by zero at $s = -1$. Since $\mathbb{E}s^2 = ab/3$, the integrated remainder is at most $12ab$.

Each $N_i$ has four monomials of coefficient one. The coefficient absolute sum of $N_1N_2N_3(1-2s)$ is at most $4^3(1 + 2 \cdot 3) = 448$. Its only pure terms are $x_1x_2x_3$ and $y_1y_2y_3$. Any mixed monomial of degree one in either vector has zero expectation by symmetry and independence. Every remaining mixed monomial has degree at least two in each vector, so its absolute expectation is at most $ab$ by the monomial bound following (2.6). The polynomial expectation differs from $\alpha + \beta$ by at most $448ab$. Adding the remainder and using the triangle inequality proves (5.2). □

This improvement uses the numerator's support constraints. A bound on $Z^{-2}$ alone cannot prove it. It also explains precisely how the private inverse-integrability warning is resolved here: the weighted numerator vanishes fast enough at the boundary. The theorem gives a lower bound for $m$, not the subadditive upper bound in MS; it does not close the companion obligation.

# 6 A degree-independent corrected potential

**Lemma 6.1** (Subadditivity after every merge, assuming MS)**.** *Assume MS and let* $0 < K \le c/3$. *Define* $f_K(t) = t + Kt^3$ *for* $t \ge 0$. *For every* $\mu, \nu \in \mathcal{C}$,

$$f_K(m(\mu \star \nu)) \le f_K(m(\mu)) + f_K(m(\nu)). \tag{6.1}$$

*Consequently for every finite conditionally independent family in* $\mathcal{C}$, *with arbitrary unequal informations* $b_1, \dots, b_k$,

$$f_K(m_{\text{combined}}) \le \sum_{j=1}^{k} f_K(b_j). \tag{6.2}$$

*There is no bound on* $k$ *in* $K$.

*Proof.* Let $a, b$ be the two input informations, $d$ the output information, and $S = a + b$. Jensen in MS gives $0 \le d \le S - cabS$. Since $f_K'(t) \ge 1$ on $[0, \infty)$,

$$f_K(d) \le f_K(S) - (S - d) \le f_K(S) - cabS.$$

The exact polynomial identity $S^3 - a^3 - b^3 = 3abS$ now gives

$$f_K(d) \le f_K(a) + f_K(b) + (3K - c)abS \le f_K(a) + f_K(b).$$

Every partial product stays in $\mathcal{C}$, so finite induction proves (6.2). The empty product has information zero. □

The corrected variable is the same cubic potential as in the source, but it is applied to each merge before summation. The source chooses $K \le c/B^3$ after keeping only one pair's saving. Lemma 6.1 permits $K = c/3$ independently of offspring count and of branch comparability. Unlike the identical-branch all-merges estimate in the local M437 report, this statement handles arbitrary unequal inputs in the nonlinear potential needed for capacity.

# 7 Extension to locally finite heterogeneous trees

## 7.1 Domain and capacity

Let $T$ be any infinite deterministic locally finite rooted tree. Each vertex has finitely many children, but the supremum of their counts may be infinite. For every edge $e$, choose a known deterministic parameter

$$0 < \lambda_e \le \ell < 1. \tag{7.1}$$

No positive lower bound is assumed. Edge transmissions remain independent given parent spins, use (1.1) with their own parameters, and the root is uniform. The observer knows both tree and parameters. Define

$$r_e = \lambda_e^2, \quad R_\rho = 1, \quad R_v = \prod_{e\in[\rho,v]} r_e^{-1}, \quad V(\theta,\xi) = \sum_{v\in\xi\setminus\{\rho\}} (R_v\theta(v))^2. \tag{7.2}$$

Here $\theta(v)$ is entering flow; define $\mathrm{Cap}_3(T;R)$ by (1.4) with this potential. It is finite because every positive-flow edge continues along an infinite ray, whence $\theta(\rho,w) \le r_{\rho w}$ and

$$|\theta| \le \sum_{w\in\mathrm{ch}(\rho)} r_{\rho w} < \infty. \tag{7.3}$$

**Theorem 7.1** (Conditional heterogeneous capacity criterion)**.** *Under MS, for every tree and edge assignment satisfying* (7.1)*,*

$$a_\infty(T,(\lambda_e)) > 0 \quad\Longleftrightarrow\quad \mathrm{Cap}_3(T;R) > 0. \tag{7.4}$$

*In particular, the assigned homogeneous theorem extends from bounded-degree trees to all locally finite trees, under exactly the same companion input.*

## 7.2 Necessity with no uniform degree bound

*Proof of necessity in Theorem 7.1.* For each vertex define $m_v$ as the decreasing limit of its descendant-level information. Finite levels are finite by local finiteness. Lemma 6.1 at finite depth, followed by the scalar limit through a finite child sum, gives with $K = c/3$ and $y_v = f_K(m_v)$,

$$\begin{aligned} y_v &\le \sum_w f_K(r_{vw}m_w) \\ &= \sum_w r_{vw}y_w - K\sum_w r_{vw}(1-r_{vw}^2)m_w^3 \\ &\le \sum_w r_{vw}y_w(1-\delta y_w^2), \qquad \delta = \frac{K(1-\ell^4)}{(1+9K)^3} > 0. \end{aligned} \tag{7.5}$$

We used $m_w \le 3$ and $r_{vw}^2 \le \ell^4$. Taking $z_v = \sqrt{2\delta}\, y_v$ and the same tangent bound yields

$$z_v \le \sum_w \frac{r_{vw}z_w}{\sqrt{1+z_w^2}}. \tag{7.6}$$

Choose $d_{vw}$ below the corresponding coefficient bounds with $z_v = \sum_w d_{vw}z_w$, and set $l_\rho = 1$, $l_w = l_v d_{vw}$, $\theta(v,w) = l_w z_w$. In a zero upper sum take all coefficients zero; otherwise scale all upper coefficients by the ratio of the desired sum to the upper sum. Conservation is exact. Set $g_v = R_v l_v$. Because $R_w = R_v/r_{vw}$,

$$g_w^2(1+z_w^2) \le g_v^2, \qquad (R_w\theta(v,w))^2 \le g_v^2 - g_w^2.$$

The ray sum telescopes to at most one, and the mass is $z_\rho$. If $a_\infty > 0$, then $m_\rho > 0$ by (2.3); hence the capacity is positive. Only finite child sums, not a uniform bound on their lengths, have been used. □

### 7.3 Degree-independent local losses

Let $C = 460$, $C_1 = 5C = 2300$, and $C_2 = C_1^2$. These explicit constants are deliberately conservative. Fix an admissible flow $\theta$ with mass $t > 0$, let $\Theta = \epsilon\theta$, $M = \epsilon t$, and set

$$X_v = R_v\Theta(v), \quad X_\rho = M, \quad X_v = \sum_w r_{vw}X_w, \quad Q = 1 + t^2. \tag{7.7}$$

Then $X_v \le \epsilon$ for nonroot positive-flow vertices, and $X_v = 0$ for zero-flow vertices. Each supported root-to-infinity path obeys $\sum_i X_{V_i}^2 \le Q\epsilon^2$. This uses the mass of the chosen flow rather than any uniform degree bound.

At level $n$, use the terminal noise $t_v = \sqrt{X_v/3}$ with $0 < \epsilon \le 1$. The revealing law has $m = 3$, $\tau = 1$, so now we retain the exact terminal third moment:

$$\widehat{m}_v = X_v, \qquad \widehat{h}_v = (X_v/3)^{3/2} = X_v^{3/2}/(3\sqrt{3}). \tag{7.8}$$

Under MS the finite-depth laws lie in $\mathcal{C}$, and $\widehat{m}_v \le X_v$. Put $b_w = r_{vw}\widehat{m}_w$, $h'_w = \lambda_{vw}^3\widehat{h}_w$, $H_v = \sum_w h'_w$. During a sequential merge, the partial $m$ is at most the sum of included $b_w$, hence at most $X_v$. Theorem 5.2 gives for every partial product $h \le H_v + CX_v^2$ and for the full product

$$\widehat{h}_v \le H_v + C_1X_v^2, \tag{7.9}$$

$$0 \le D_v := \sum_w b_w - \widehat{m}_v \le 10X_vH_v + 5CX_v^3 \le C_1X_v(H_v + X_v^2). \tag{7.10}$$

Indeed the quadratic errors sum to at most $C\sum_j b_j \sum_{i<j} b_i \le CX_v^2$; the quadratic-information deficits sum to at most $5\sum_j\{(H_v + CX_v^2)b_j + X_vh'_j\}$. Neither argument counts the number of children. All sums at a vertex are finite. The lower sign of $D_v$ comes from MS, not from Theorem 5.2.

### 7.4 Full path accumulation and a fixed noise scale

For an ancestor $v$ of $u$, let $\Lambda(v,u) = \prod_{e\in[v,u]}\lambda_e$, with $\Lambda(v,v) = 1$. Iterating (7.9) inside (7.10) gives

$$D_v \le C_2X_v\left\{\sum_{\substack{u\succeq v\\ |u|<n}}\Lambda(v,u)^3X_u^2 + \frac{1}{3\sqrt{3}}\sum_{\substack{u\succeq v\\ |u|=n}}\Lambda(v,u)^3X_u^{3/2}\right\}. \tag{7.11}$$

Every internal forcing term is inserted once and propagated with $\Lambda^3$; this proves the formula by finite iteration with $C_2 \ge C_1, C_1^2$. Telescoping with weights $R_v^{-1}$ yields

$$\widehat{m}_\rho = M - \sum_{|v|<n} R_v^{-1}D_v. \tag{7.12}$$

Finite branches carry zero flow, so conservation gives the terminal sum $M$ exactly.

Sample the depth-$n$ path with entering-flow marginals $\Theta(u)/M$. Since $R_u/R_v = \Lambda(v,u)^{-2}$, the crucial identity becomes

$$R_v^{-1}\Lambda(v,u)^3X_u = \Lambda(v,u)\Theta(u). \tag{7.13}$$

It turns the internal and terminal sums, respectively, into

$$M\mathbb{E}[\Lambda(V_i,V_j)X_{V_i}X_{V_j}], \qquad \frac{M}{3\sqrt{3}}\mathbb{E}[\Lambda(V_i,V_n)X_{V_i}\sqrt{X_{V_n}}].$$

By (7.1), $\Lambda(V_i, V_j) \le \ell^{j-i}$. For each lag $0 \le k < n$,

$$\sum_{i=0}^{n-1-k} X_{V_i} X_{V_{i+k}} \le \sum_{i=0}^{n} X_{V_i}^2 \le Q\epsilon^2$$

where the first displayed inequality is understood by Cauchy–Schwarz on the two truncated sums. Also $X_{V_i} \le \sqrt{Q}\,\epsilon$ and $X_{V_n} \le \epsilon$. Thus, uniformly for every $n \ge 1$,

$$\frac{1}{M} \sum_{|v|<n} R_v^{-1} D_v \le \frac{C_2}{1-\ell} \left( Q\epsilon^2 + \frac{\sqrt{Q}\,\epsilon^{3/2}}{3\sqrt{3}} \right). \tag{7.14}$$

Compared with the assigned proof's terminal $O(\epsilon)$ loss, retaining (7.8) gives $O(\epsilon^{3/2})$ for a fixed chosen flow. This is a bound on the proof's degradation loss, not a sharp reconstruction exponent.

Define the explicit number

$$\alpha_\ell = \min \left\{ 1, \left( \frac{1-\ell}{4C_2} \right)^{1/2}, \left( \frac{3\sqrt{3}(1-\ell)}{4C_2} \right)^{2/3} \right\} > 0, \quad \epsilon = \alpha_\ell / \sqrt{1+t^2}. \tag{7.15}$$

Then $Q\epsilon^2 = \alpha_\ell^2$ and $\sqrt{Q}\epsilon^{3/2} \le \alpha_\ell^{3/2}$. Each term in (7.14) contributes at most $1/4$. Hence $\widehat{m}_\rho \ge M/2$ for all $n$. Data processing and (2.3) give

$$a_n \ge \frac{\alpha_\ell}{16} \frac{t}{\sqrt{1+t^2}} > 0 \qquad (n \ge 1). \tag{7.16}$$

This proves sufficiency and completes Theorem 7.1. The scale is independent of observation depth; no lower edge bound was used.

**Corollary 7.2** (Quantitative comparison, assuming MS)**.** *Write* $\mathsf{C} = \mathrm{Cap}_3(T; R)$ *and let* $\delta$ *be as in* (7.5)*. Then*

$$\frac{\alpha_\ell}{16} \frac{\mathsf{C}}{\sqrt{1+\mathsf{C}^2}} \le a_\infty \le \frac{\sqrt{\mathsf{C}}}{2(2\delta)^{1/4}}. \tag{7.17}$$

*Proof.* For the lower bound take admissible masses tending to the finite supremum in (7.16); attainment is unnecessary. For the upper bound the constructed flow gives $\mathsf{C} \ge \sqrt{2\delta} f_K(m_\rho) \ge \sqrt{2\delta} m_\rho$, and (2.3) gives $a_\infty \le \sqrt{m_\rho}/2$. □

These constants depend only on $\ell$ and the fixed MS constant, not on a uniform degree bound. They degenerate as $\ell \uparrow 1$. The theorem does not cover infinitely many children at a vertex, unknown or spin-dependent edge parameters, antiferromagnetic channels, or arbitrary assignments approaching one with no common upper bound. The latter may be approachable through path-kernel estimates but is not proved here.

## 8 Finite capacities, errors, and critical examples

This section is deterministic and independent of MS. It describes what a finite computation can and cannot establish about the actual nonlinear capacity.

**Proposition 8.1** (Finite recursion and infinite limit)**.** *For the tree truncated at level $N$, let $c_v^{(N)}$ be its capacity from $v$ to its level-$N$ descendants, with resistances restarted at $v$. Use $c_v^{(N)} = \infty$ at depth $N$ and $0$ at earlier dead leaves. Then, for $|v| < N$,*

$$c_v^{(N)} = \sum_{w \in \mathrm{ch}(v)} r_{vw} \frac{c_w^{(N)}}{\sqrt{1 + (c_w^{(N)})^2}}, \qquad \frac{\infty}{\sqrt{1+\infty^2}} := 1. \tag{8.1}$$

*Moreover $c_\rho^{(N)} \downarrow \mathrm{Cap}_3(T; R)$.*

*Proof.* For a branch through $w$, an entering mass $b$ uses $(b/r_{vw})^2$ potential on its first edge and at least $(b/(r_{vw}c_w^{(N)}))^2$ in its remaining subtree. Optimizing its allowed mass gives $r_{vw}c_w^{(N)}/\sqrt{1+(c_w^{(N)})^2}$. The limiting conventions at zero and infinity follow directly. Distinct branches have disjoint rays, so their capacities add. Induction proves (8.1). This is the $p = 3$ classical capacity recursion [3] with our explicit terminal convention.

Restricting an infinite admissible flow to depth $N$ gives $\mathrm{Cap}_3 \le c_\rho^{(N)}$, and restricting a deeper finite flow shows monotone decrease. For the reverse inequality, choose finite admissible flows whose masses approach the decreasing limit. For each fixed edge their masses are in $[0, R_v^{-1}]$ whenever the edge belongs to a positive path to the terminal level. A diagonal subsequence exists over the countable edge set. Local finiteness allows passage of conservation and root mass through finite sums. Every finite path potential is at most one in the limit. Monotone convergence along an infinite ray then gives an admissible infinite flow of the limiting mass. Earlier leaves have zero incoming flow. Thus the limit equals capacity. □

**Proposition 8.2** (A residual budget for finite evaluation)**.** *Let $\widetilde{c}_v \ge 0$ use the exact depth-$N$ terminal convention and equal zero at earlier dead leaves. At nonleaf vertices $|v| < N$, bound the absolute residual in* (8.1) *by $e_v$. Then*

$$|\widetilde{c}_\rho - c_\rho^{(N)}| \le \sum_{|v|<N} R_v^{-1} e_v. \tag{8.2}$$

*Set $e_v = 0$ at earlier dead leaves. An independently supplied finite child-label error is weighted by the product of $r_e$ on its root path.*

*Proof.* For $x \ge 0$, $g_e(x) = r_e x/\sqrt{1+x^2}$ has derivative $r_e(1+x^2)^{-3/2} \le r_e$. Subtract the two recursions to get $\eta_v \le e_v + \sum_w r_{vw}\eta_w$. Iterate to the exact terminal layer. Each ancestor chain has product $R_v^{-1}$, proving the bound. The infinite terminal symbols are evaluated exactly through $g_e(\infty) = r_e$, not subtracted as infinities. □

Interval evaluation or exact square-root enclosures can therefore bound a finite capacity. It is an *upper* bound on the infinite capacity. A positive finite value, or even many such values, is not a positive lower bound on their limit. A global lower certificate needs a complete admissible flow of positive mass or a valid infinite-tail argument. A computed zero within numerical tolerance is not an exact zero certificate either.

**Proposition 8.3** (Spherical specialization and separation from electricity)**.** *Suppose every depth-$n$ vertex has the same number of children and all edges entering level $n$ have the same parameter. Let $N_n = |L_n| > 0$, $R_n = \prod_{j=1}^n \lambda_j^{-2}$. Then*

$$\mathrm{Cap}_3(T; R) = \left[\sum_{n=1}^{\infty} (R_n/N_n)^2\right]^{-1/2}, \tag{8.3}$$

*where a divergent sum gives zero. For ordinary electrical $L^2$ capacity with the* same *resistances, the corresponding expression is* $[\sum_n R_n/N_n]^{-1}$.

*Proof.* The equally splitting flow of mass $t$ has entering mass $t/N_n$ at depth $n$, so every ray has potential $t^2 \sum_n (R_n/N_n)^2$. Conversely, starting at the root, select at each step a child carrying at least the average outgoing flow. Spherical branching ensures entering mass at depth $n$ at least $t/N_n$ along this one infinite ray. Thus no mass-$t$ flow has smaller supremum potential. The same proof with potential exponent one gives the electrical formula. This is a direct specialization of classical capacity theory, not a new capacity identity. □

For a $d$-ary tree and $r = \lambda^2$, this yields zero capacity when $dr \le 1$ and $\sqrt{(dr)^2 - 1}$ when $dr > 1$. At critical exponential growth, if $N_n r^n$ is comparable to $n^\gamma$, the $L^3$ criterion is $\sum n^{-2\gamma} < \infty$, equivalently $\gamma > 1/2$. An electrical substitution instead demands $\gamma > 1$.

An explicit bounded-degree example uses $r = 1/2$, $1/2 < \gamma \le 1$, and

$$N_n = 2^n 2^{\lfloor \gamma \log_2(n+1) \rfloor}.$$

The ratio $N_n/N_{n-1}$ is two or four, so this is realized by a spherical tree with at most four children. The floor changes only bounded factors. Its nonlinear capacity is positive while its electrical capacity with resistances $2^n$ is zero. Under MS it reconstructs by the assigned theorem itself. This example illustrates the inherited theorem; it is not credited as the report's extension.

On an infinite ray $c_\rho^{(N)} = [\sum_{n=1}^N r^{-2n}]^{-1/2} > 0$ for every finite $N$, but the limit is zero for $r < 1$. At $r = 1$ the finite values are $N^{-1/2}$ while a noiseless spin reconstructs perfectly. These exact examples simultaneously test the terminal convention, the infinite-limit boundary, and the excluded endpoint.

# 9 Verification, advantages, and remaining work

## 9.1 Bounded mathematical diagnostics

The accompanying author-owned `evidence/checks.py` completed 60 grouped checks with zero failures. It uses exact rational or symbolic arithmetic for the polynomial identities, and separately marked 70-digit non-interval arithmetic for square-root recursions. It checks the coordinate inverse and all color actions, the moment cancellation and denominator identities, the coefficient norm used in Theorem 5.2, the corrected-potential identity, selected exact symmetric posterior products (including zero denominators and a negative signed third moment), a rational heterogeneous flow-weight identity, the finite Gram counterexample, and finite capacity/residual fixtures. Its JSON receipt records every test's scope and result.

The proofs above, not a finite posterior sample, justify the all-law and all-tree quantifiers. In particular the diagnostics do not replay either companion traversal, validate arbitrary empirical posterior laws, prove MS, or establish a continuum statement by dense sampling. The stored build log, read scopes, SHA-256 input identities, final-page inspection record and frozen manifest keep scientific claims distinct from document-generation checks.

## 9.2 Advantages and costs

The useful Ram formulation preserves one probability law per contraction, the distinction between marginal independence and Bayes tilting, and the exact signed moment responsible for the four-state

cancellation. Polynomial synthesis allows independently checkable algebra. Paying for the full rational remainder exposes cancellations invisible to a reciprocal-only bound. The cubic potential turns each pairwise saving into a degree-independent inequality, while the path representation prevents artificial constants from accumulating exponentially with depth.

The method has real limitations. Finite posterior moments do not determine a rational Bayes update. A covariance matrix can be PSD without being realizable on the tetrahedron. Dense reference Gram transforms may be more expensive and less well conditioned than sparse coefficient arithmetic. The explicit constants in this report are coarse and may give very small positive advantage bounds. The global theorem still rests on MS, and finite capacity computation alone only gives decreasing upper bounds. No benchmark establishes faster verification or stronger practical reconstruction.

### 9.3 Specific future directions

The highest-priority missing evidence is an independent assertion-enabled replay of both unchanged companion certificate modes, together with the complete accepted-leaf data and split coverage. That would address the shared input rather than merely adding more sample pairs. A small separate checker should verify the exported coefficient arrays and genealogy.

The constant 460 can be reduced by eliminating symmetry-zero monomials before bounding the polynomial; the exact algebra is small enough to investigate, but an optimized value is not needed for the extension. Better constants in the nonlinear information/capacity comparison may require a different potential or a sharper terminal design, and would need quantitative consequences beyond numerical-factor changes.

For edge assignments with no uniform $\ell < 1$, the path kernel $\Lambda(V_i, V_j)$ suggests a replacement condition in terms of uniform summability or a weighted Schur bound. Necessity also requires control of the loss $1 - r_e^2$ in the corrected potential. Controlling one side alone does not prove a two-sided theorem. Infinite offspring at a single vertex would additionally require convergence of products and sums, not just a relabeling of local finiteness.

Finite computations could become useful infinite lower certificates by appending verified subtree capacity bounds or an explicit tail flow. Such tail assumptions must be stated and proved for the actual tree. The regular, spherical, and finite-depth formulas in this report are test cases for that work, not universal tail certificates.

## 10 Reproducibility and assessment

The read-scope ledger records the complete assigned main source and bibliography/README; the original companion Sections 2, 3, 5 and 6 plus its checks README; the complete related local M437 report; all three routing maps; catalogue row 438; the reused upstream findings; and the selected actual private proofs listed in Section 4. It does not claim a full read or execution of the companion C++ appendix, a replay of private numerical suites, or a visual review of the source PDF. No original or shared catalogue/queue/progress file was changed.

The public literature record identifies the inspected primary URLs and version information. The Pemantle author-host PDF endpoint failed; the arXiv v3 PDF supplied the definitions and relevant theorem/recursion passages. Sly and Janson–Mossel were inspected at abstract/metadata level; Mossel–Sly–Sohn at abstract and main broadcast-theorem level. Searches used no confidential passages or private filenames.

The realized assessment is C3, separately from L4 applicability. The reasons are Theorem 5.2, Lemma 6.1, Theorem 7.1, and Corollary 7.2; the sharper terminal loss is part of their quantitative proof. Classical finite-capacity identities, exact coordinate transport, source reconstruction results and unreplayed certificates are not counted as new contributions. Removing the bounded-degree hypothesis is a mathematical consequence under the named premise; it is not a claim that the companion premise has been discharged or that historical priority has been established.

## References

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[2] OpenAI. *The Reconstruction Threshold for the Ferromagnetic Four-State Potts Model*. October 5, 2026, same pinned repository. Sections 2, 3, 5 and 6, especially `prop:cone-closure`, `prop:polynomial-inequalities` and `prop:finite-verification`. Complete computational certification is source-reported.

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